Analysis of Student Performance Factors: *What Drives Student Success*

OMIS 630 – Python Programming for Business

Group Number: 5

Group Members: Akshata Pai (Z2069546) & Tanvi Aggarwal (Z2079466)

1. Data Source & Description

1.1 Dataset Overview

The Student Performance Factors dataset (source: Kaggle) contains structured data designed to analyze the various factors that influence students' academic outcomes, particularly their final exam scores. The dataset consists of multiple observations, where each row represents an individual student and each column corresponds to a specific attribute related to academic behaviour, personal habits, or socioeconomic background.

The variables included in the dataset capture a wide range of influences on student performance. These can be broadly categorized into three groups: academic factors (such as study hours, attendance, previous scores, and tutoring sessions), lifestyle factors (including sleep patterns, physical activity, and extracurricular involvement), and environmental or socioeconomic factors (such as family income, parental education, school type, and access to resources like the internet). The primary outcome variable is the student's exam score, which serves as the target for analysis and prediction.

This dataset is synthetic, meaning it has been generated for educational and analytical purposes rather than collected from real-world student records. As such, it allows researchers and students to explore relationships between variables, perform exploratory data analysis, and build predictive models without concerns related to privacy or sensitive data. Overall, it provides a comprehensive framework for studying how different behavioural, personal, and contextual factors contribute to student achievement.

1.2 Dataset Dimensions

- Number of observations (rows): 6607
- Number of variables (columns):20
- Types of variables:
    - Numerical (Hours Studied, Attendance, Sleep_Hours, Previous Scores, Tutoring_Sessions, Physical Activity, Exam Score)
    - Categorical (Parental_Involvement, Access_to_Resources, Extracurricular_Activities, Motivation Level, Internet_Access Family_Income,

Teacher_Quality, School_Type, Peer_Influence, Learning_Disabilities, Parental_Education_Level, Distance_from_Home, Gender)

2. Python Code

- Link to **Google Collab Notebook:**

  https://colab.research.google.com/drive/1J5uEYaPQuef8UU69xohJW1lYVlX7hglL?usp=sharing

3. Exploratory Data Analysis (EDA)

3.1 Dataset Structure

The dataset consists of 6,607 student records with a total of 20 variables, capturing a wide range of academic, behavioural, and socioeconomic factors that may influence student performance. Each row represents an individual student, and the dataset includes both numerical and categorical features.

Out of the 20 variables:

7 are numerical (integer type), representing measurable quantities such as study hours and exam scores.

13 are categorical (object type), representing qualitative attributes such as parental involvement, access to resources, and school type.

Most variables contain complete data; however, a few columns such as *Teacher_Quality*, *Parental_Education_Level*, and *Distance_from_Home* contain some missing values, indicating minor data incompleteness that required preprocessing. As all the 3 variables were categorical in nature, we use Mode Imputation with FILLNA to backfill the missing values.

```
student.isna().sum()

# Here we can see number of missing values
# Teacher_Quality has 78 missing values, Parental_Education_Level has 90
missing values and
# Distance_from_Home has 67 missing values
```

	0
Hours_Studied	0
Attendance	0
Parental_Involvement	0
Access_to_Resources	0
Extracurricular_Activities	0
Sleep_Hours	0
Previous_Scores	0
Motivation_Level	0
Internet_Access	0
Tutoring_Sessions	0
Family_Income	0
Teacher_Quality	78
School_Type	0
Peer_Influence	0
Physical_Activity	0
Learning_Disabilities	0
Parental_Education_Level	90
Distance_from_Home	67
Gender	0
Exam_Score	0

dtype: int64

```
Distance_from_Home_mode = student["Distance_from_Home"].mode()[0]
print(Distance_from_Home_mode)

student["Distance_from_Home"] = student["Distance_from_Home"].fillna
(Distance_from_Home_mode)
display(student)
```

Near

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions	Family_Income	Teacher_Quality	Schoo
0	23	84	Low	High	No	7	73	Low	Yes	0	Low	Medium	
1	19	64	Low	Medium	No	8	59	Low	Yes	2	Medium	Medium	
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	2	Medium	Medium	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	1	Medium	Medium	
4	19	92	Medium	Medium	Yes	6	65	Medium	Yes	3	Medium	High	
...	...	...	...	...	...	...	...	...	...	...	...	...	
6602	25	69	High	Medium	No	7	76	Medium	Yes	1	High	Medium	
6603	23	76	High	Medium	No	8	81	Medium	Yes	3	Low	High	
6604	20	90	Medium	Low	Yes	6	65	Low	Yes	3	Low	Medium	
6605	10	86	High	High	Yes	6	91	High	Yes	2	Low	Medium	
6606	15	67	Medium	Low	Yes	9	94	Medium	Yes	0	Medium	Medium	

6607 rows × 20 columns

3.2 Key Variables Description

The dataset includes key variables such as:

- Academic factors: Hours_Studied, Attendance, Previous_Scores, Tutoring_Sessions, Exam_Score
- Lifestyle factors: Sleep_Hours, Physical_Activity
- Socioeconomic and environmental factors: Family_Income, Parental_Involvement, Access_to_Resources, Internet_Access, School_Type
- Other contextual variables: Motivation_Level, Peer_Influence, Learning_Disabilities, Gender

The target variable for analysis is Exam_Score, which reflects student academic performance.

Descriptive Statistics

The summary statistics provide insights into the distribution and variability of key numerical features:

Academic Performance

- The average exam score is approximately 67.24, with a standard deviation of 3.89, indicating relatively low variability in final scores.
- Scores range from 55 to 101, suggesting most students perform within a moderate band, with a few high-performing outliers.

Study Behaviour

- Students study an average of ~20 hours, with values ranging from 1 to 44 hours, indicating significant variation in study effort.
- The interquartile range (16–24 hours) shows that most students fall within a moderate study range.

Attendance

- The average attendance rate is approximately 80%, with a relatively widespread (60% to 100%).
- This suggests attendance varies considerably and may be an important factor influencing performance.

Sleep Patterns

- Students report an average of ~7 hours of sleep, ranging from 4 to 10 hours, indicating generally healthy sleep habits with some variation.

Previous Academic Performance

- The mean previous score is approximately 75, which is higher than the current average exam score (~67), suggesting a potential decline in overall performance across the dataset.

Tutoring and Physical Activity

- Students attend approximately 1.5 tutoring sessions on average, with some receiving up to 8 sessions.
- Physical activity averages around 3 units, indicating moderate engagement in non-academic activities.

Overall, the dataset provides a comprehensive view of student performance by integrating academic, behavioural, and contextual factors. The combination of

diverse variables, relatively complete data, and meaningful variation across features makes it highly suitable for regression, classification, and clustering analyses aimed at understanding and predicting student success.

4. Data Visualization

Data visualization was utilized as an essential exploratory tool to examine the distribution, relationships, and interactions among key variables influencing student performance. Through visual analysis, complex patterns and hidden trends can be identified, supporting more informed modelling and decision-making.

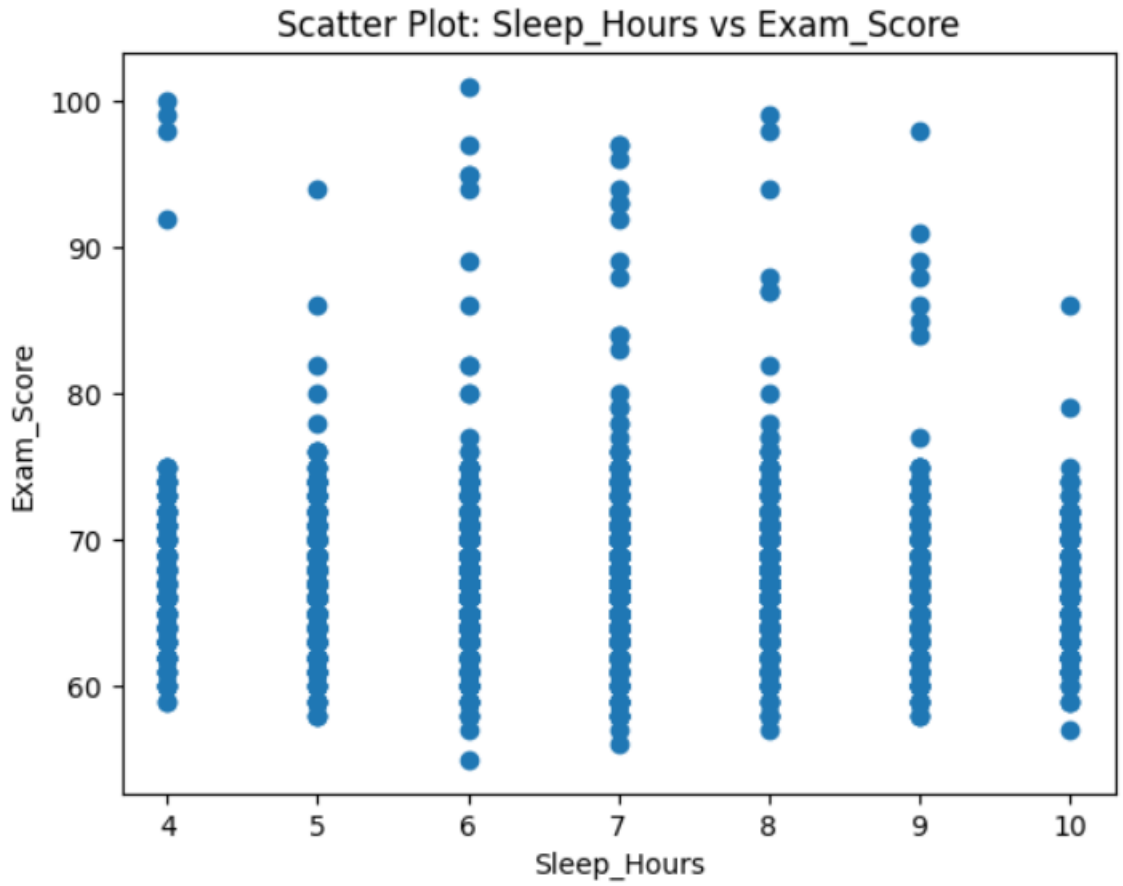
4.1 Relationship Between Sleep Duration and Student Exam Performance

```
# create a scatter plot
plt.scatter(student["Sleep_Hours"], student["Exam_Score"])

# add labels and title
plt.xlabel("Sleep_Hours")
plt.ylabel("Exam_Score")
plt.title("Scatter Plot: Sleep_Hours vs Exam_Score")

# display the chart
plt.show()
```

The scatter plot illustrates the relationship between students' sleep duration (Sleep_Hours) and their exam performance (Exam_Score). Each point in the plot represents an individual student, with sleep hours on the x-axis and exam scores on the y-axis.



From the visualization, it is evident that students typically sleep between 4 to 10 hours, with a noticeable concentration around 6 to 8 hours, indicating this is the most common sleep range. Exam scores, on the other hand, are distributed between approximately 55 and 100, with the majority clustering between 60 and 75.

The overall pattern suggests a weak positive relationship between sleep duration and exam performance. Students who sleep within the moderate range (6–8 hours) tend to achieve relatively consistent and slightly higher scores compared to those with very low sleep durations. However, the relationship is not strongly linear, as there is significant overlap in exam scores across all sleep levels.

Additionally, students with both lower (4–5 hours) and higher (9–10 hours) sleep durations display a wide spread of performance outcomes, indicating that sleep alone is not a dominant predictor of exam success. High-performing students are observed across multiple sleep categories, suggesting that other factors such as study habits, attendance, and prior academic performance play a more substantial role.

While adequate sleep appears to support stable academic performance, the scatter plot highlights that optimal performance is influenced by a combination of factors, and sleep duration should be considered as one contributing element rather than a sole determinant.

4.2 Impact of Parental Involvement on Exam Performance Across Sleep Duration

The line chart presents the relationship between sleep duration and average exam scores, segmented by levels of parental involvement (Low, Medium, and High). The data is aggregated using a pivot table, where the mean exam score is calculated for each combination of sleep hours and parental involvement category.

```
# Prepare the data (Pivot Table)
AvgExamScore = student.pivot_table(index='Sleep_Hours',
                                   columns='Parental_Involvement',
                                   values='Exam_Score',
                                   aggfunc='mean')

# create a line chart
plt.plot(AvgExamScore.index, AvgExamScore.values)

# add labels and title
plt.xlabel("Sleep hours")
plt.ylabel("Average Exam Score")
plt.title("Average Exam Score by Parental Involvement Over Hours Studied")

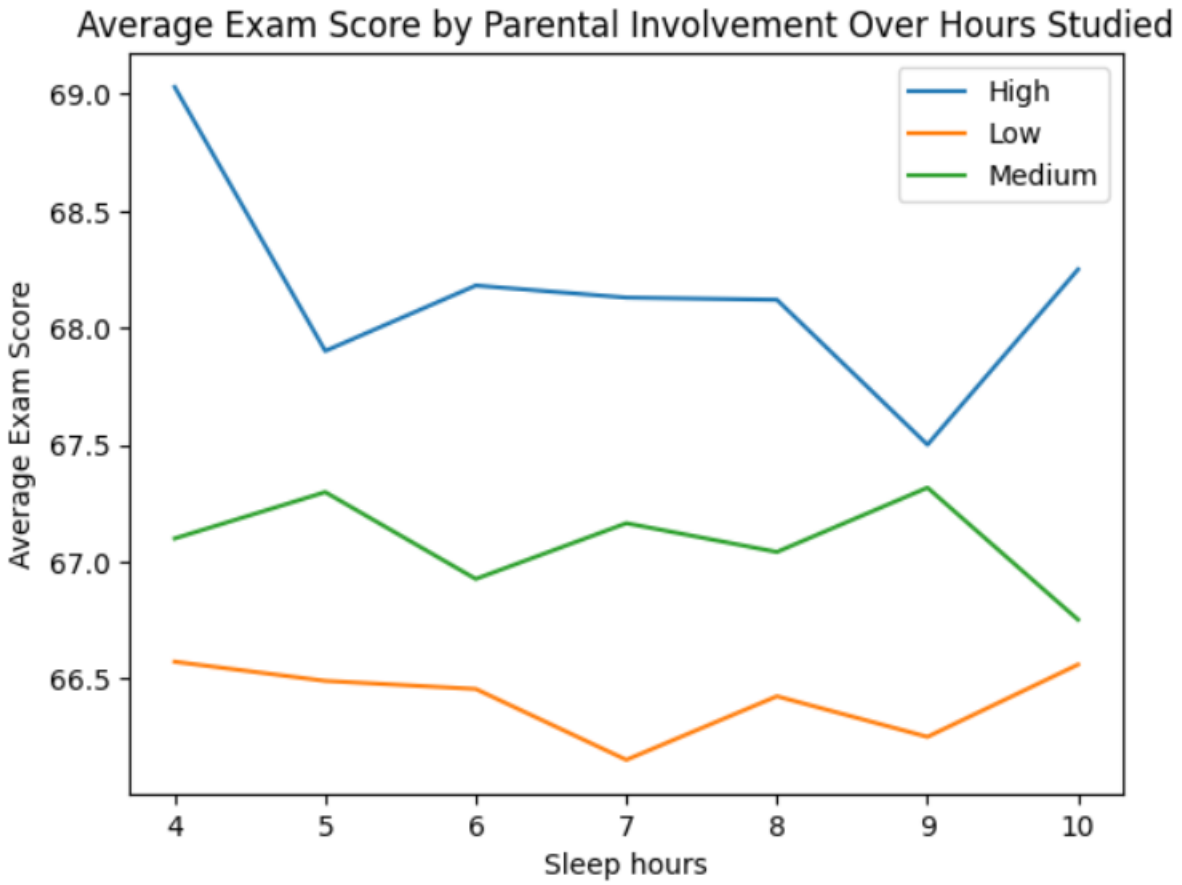
# ensure the value on x-axis is the value of pivot.index
plt.xticks(AvgExamScore.index)

# add color legend
plt.legend(AvgExamScore.columns)

# display the chart
plt.show()

## Here we see students with high parental involvement generally score higher
than low and medium involvement
```

From the visualization, a clear pattern emerges students with high parental involvement consistently achieve higher average exam scores across all sleep durations compared to those with medium or low involvement. This suggests that parental engagement plays a significant role in supporting academic success, regardless of sleep patterns.



Students with medium parental involvement demonstrate moderate performance, consistently scoring above those with low involvement but below the high-involvement group. Meanwhile, students with low parental involvement tend to have the lowest average scores across nearly all sleep levels, indicating comparatively weaker academic outcomes.

In terms of sleep duration, exam performance across all groups shows minor fluctuations rather than a strong upward or downward trend. Scores remain relatively stable within the 6–8-hour range, suggesting this may be an optimal or balanced sleep duration. However, variations at extreme sleep levels (e.g., 4 or 10 hours) do not significantly alter the overall ranking between the parental involvement groups.

The chart highlights that parental involvement is a stronger and more consistent differentiator of academic performance than sleep duration alone. While adequate sleep contributes to maintaining stable performance, higher levels of parental engagement are associated with better academic outcomes across all sleep categories.

4.3 Distribution of Students by Gender Across Levels of Parental Involvement

The bar chart illustrates the distribution of students by gender (Male and Female) across different levels of parental involvement (Low, Medium, and High). The data is

aggregated using a pivot table, where the count of students is calculated for each combination of gender and parental involvement category.

```
CountData = pd.pivot_table(
    student,
    index = "Parental_Involvement",
    columns = "Gender",
    values = "Exam_Score",
    aggfunc = "count"
).plot(kind = "bar")

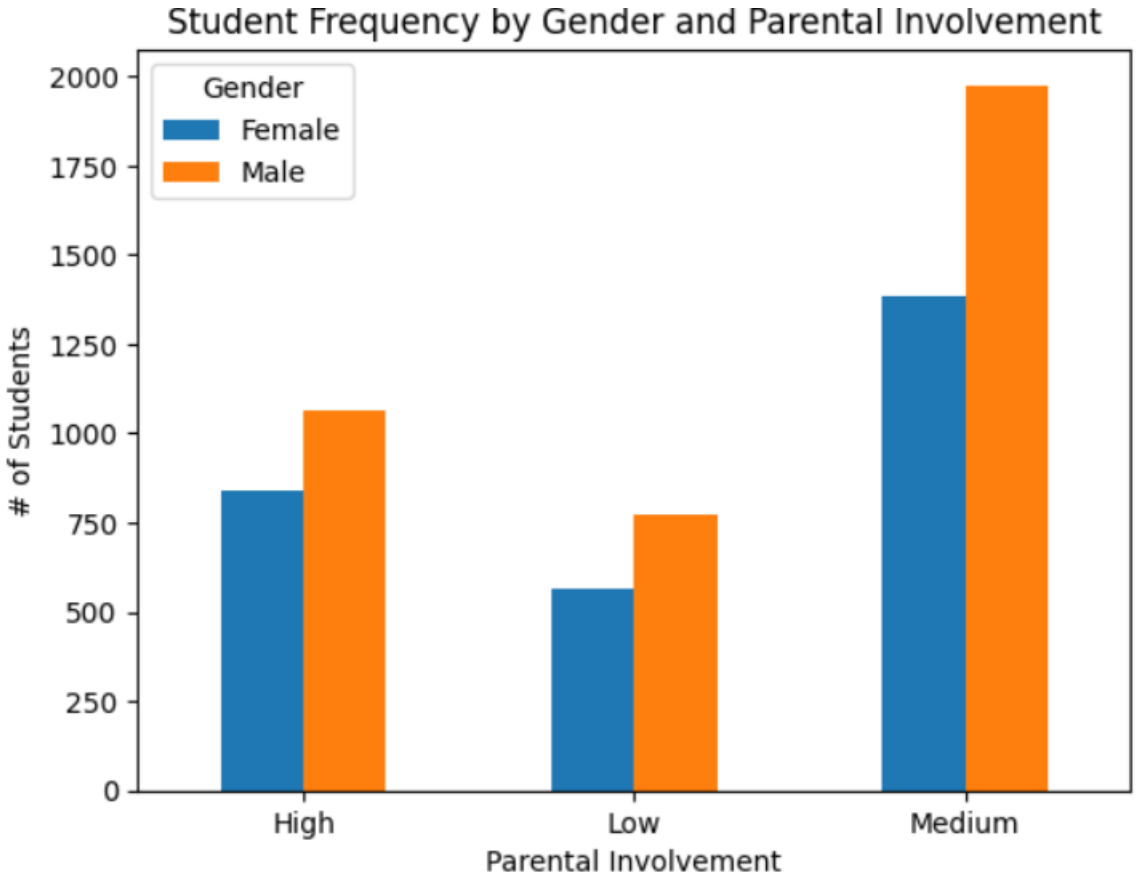
# add labels and title
plt.xlabel("Parental Involvement")
plt.ylabel("# of Students")
plt.title("Student Frequency by Gender and Parental Involvement")

# ensure the labels on x-axis are horizontal
plt.xticks(rotation=0)

# display the chart
plt.show()

## Across all levels of parental involvement, number of male students are
higher than female students
```

From the visualization, it is evident that male students outnumber female students across all levels of parental involvement. This pattern is consistent in the Low, Medium, and High categories, indicating a gender imbalance in the dataset.



The Medium parental involvement category contains the largest number of students for both genders, suggesting that most students experience a moderate level of parental engagement. In contrast, the Low involvement category has the smallest number of students, indicating that relatively fewer students fall into this group.

Although the counts differ, the distribution trend across parental involvement levels is similar for both genders. Both male and female students follow the same pattern: Medium involvement is most common, followed by High, and then Low. This suggests that parental involvement levels are not disproportionately skewed toward one gender in terms of relative distribution, even though absolute counts differ.

While there is a clear numerical dominance of male students in the dataset, the pattern of parental involvement is consistent across genders, indicating that gender does not significantly influence the distribution of parental engagement levels. This suggests that any differences in academic performance related to parental involvement are likely independent of gender distribution patterns.

5. Machine Learning Analysis

5.1 Technique 1 (Multiple Regression)

Objective

To develop and evaluate regression models that accurately predict student exam performance using study habits, prior academic history, and related influencing factors.

Analytical Question

To what extent and with what level of accuracy can student exam performance be predicted using study habits, prior academic history, and other relevant variables through regression models?

Methodology:

A multiple linear regression approach was employed to model and predict student exam performance. The Ordinary Least Squares (OLS) method was implemented using the statsmodels library, with *Exam_Score* specified as the dependent variable. Independent variables included key academic and behavioural factors such as hours studied, attendance, sleep duration, tutoring sessions, physical activity, and previous academic scores.

In addition, several categorical variables parental involvement, access to resources, motivation level, and internet access were incorporated into the model using appropriate encoding to capture their effects. The model was fitted to the dataset, and statistical outputs such as coefficients, significance levels, and goodness-of-fit metrics were analysed to evaluate the strength, direction, and predictive capability of the relationships between variables and exam performance.

```
import statsmodels.formula.api as smf

model = smf.ols("Exam_Score ~ Hours_Studied + Attendance + Sleep_Hours+
Tutoring_Sessions + Physical_Activity + Previous_Scores + C
(Parental_Involvement) + C(Access_to_Resources)+ C(Motivation_Level)+ C
(Internet_Access)", data=df).fit()

print(model.summary())
```

We ran the model twice in first iteration we discovered that Sleep Hours was not significant predictor of the exam scores and hence we dropped sleep hours from our final model and re-ran our regression model.

Final Model:

```
## Sleep Hours is not significant predictor of Exam scores as the p-value (0.
628) > 0.05 hence we remove it from our model and re-run the model.
## Final Model

import statsmodels.formula.api as smf

model = smf.ols("Exam_Score ~ Hours_Studied + Attendance + Tutoring_Sessions +
Physical_Activity + Previous_Scores + C(Parental_Involvement) + C
(Access_to_Resources)+ C(Motivation_Level)+ C(Internet_Access)", data=df).fit()


print(model.summary())
```

Results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            Exam_Score   R-squared:                       0.675
Model:                           OLS   Adj. R-squared:                  0.674
Method:                Least Squares   F-statistic:                     1139.
Date:               Mon, 27 Apr 2026   Prob (F-statistic):               0.00
Time:                       01:42:07   Log-Likelihood:                -14642.
No. Observations:               6607   AIC:                         2.931e+04
Df Residuals:                   6594   BIC:                         2.940e+04
Df Model:                         12
Covariance Type:            nonrobust
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                       42.1618      0.300    140.359      0.000      41.573      42.751
C(Parental_Involvement)[T.Low]  -2.0135      0.079    -25.375      0.000      -2.169      -1.858
C(Parental_Involvement)[T.Medium]  -1.0478      0.064    -16.448      0.000      -1.173      -0.923
C(Access_to_Resources)[T.Low]   -2.0419      0.079    -25.779      0.000      -2.197      -1.887
C(Access_to_Resources)[T.Medium]  -0.9621      0.063    -15.227      0.000      -1.086      -0.838
C(Motivation_Level)[T.Low]      -1.0681      0.079    -13.459      0.000      -1.224      -0.912
C(Motivation_Level)[T.Medium]   -0.5472      0.072     -7.571      0.000      -0.689      -0.406
C(Internet_Access)[T.Yes]        0.8834      0.104      8.534      0.000       0.680       1.086
Hours_Studied                    0.2947      0.005     64.519      0.000       0.286       0.304
Attendance                       0.1998      0.002     84.341      0.000       0.195       0.204
Tutoring_Sessions                0.5026      0.022     22.612      0.000       0.459       0.546
Physical_Activity                0.1592      0.027      6.000      0.000       0.107       0.211
Previous_Scores                  0.0478      0.002     25.124      0.000       0.044       0.051
==============================================================================
Omnibus:                     9620.593   Durbin-Watson:                   2.013
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          2804592.225
Skew:                           8.918   Prob(JB):                         0.00
Kurtosis:                     102.346   Cond. No.                     1.25e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.25e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Statistical Perspective

Exam_Scores = 0.2947 * (Hours_Studied) + 0.1998 * (Attendance) + 0.5026 * (Tutoring_Sessions) + 0.1592 * (Physical_Activity) + 0.0478 (Previous_Scores) -2.0135 * (Low_Parental_Involvement) - 1.0478 (Medium_Parental_involvement) -2.0419 * (Low_Access_to_Resources) -0.9621 * (Medium_Access_to_Resources) -1.0681

*(Low_Motivation_Level) -0.5472 (Medium_Motivation_Level) + 0.8834 (Yes_Internet_Access)

Interpreting The R-Square (Model fit): About 67.4% of the variation in Exam_Scores is explained by the model.

As the p-value of the F-statistics (0.00) is less than 0.05, We have a significant model.

Hours_Studied, Attendance, Tutoring_Sessions among other variables are significant predictors of Exam_Scores.

Holding all other variables constant, the average exam score is expected to be 42.23 when all predictor variables are equal to zero.

A one-unit increase in study hours is associated with an average increase of approximately 0.295 points in the exam score.

A one-unit increase in attendance is associated with an average increase of approximately 0.1998 points in the exam score.

A one-unit increase in tutoring-sessions is associated with an average increase of approximately 0.5026 points in the exam score.

A one-unit increase in the physical-activity is associated with an average increase of approximately 0.1592 points in the exam score.

A one-unit increase in previous exam scores is associated with an average increase of approximately 0.047 points in the current exam score.

Having Internet Access is associated with 0.8834 points increase in the exam score

Compared to “High” involvement, lower involvement reduces scores.

Higher the Motivation better the scores

Lower the access to resources lower the scores.

Practical Perspective

Student exam performance can be predicted to a substantial and meaningful extent using study habits, prior academic history, and related factors through regression models.

The model explains approximately 67% of the variation in exam scores ($R^2 \approx 0.67$), which indicates a good level of predictive accuracy for an educational dataset. This means that key variables such as study hours, attendance, tutoring sessions, and prior performance collectively provide a reliable basis for estimating student outcomes.

From a practical standpoint:

Study behaviour variables (e.g., hours studied, attendance, tutoring) have the strongest and most consistent impact on performance.

Prior academic performance contributes positively, reflecting baseline ability, though its effect is relatively smaller.

Contextual factors (such as parental involvement, access to resources, and motivation) also play a significant role in shaping outcomes.

Importantly, the absence of multicollinearity (all VIF < 3) suggests that these factors independently contribute to prediction, making the model stable and interpretable.

In Real-life, Educators can reasonably predict student performance using observable behaviours and past records.

The model is accurate enough to support: Early academic monitoring Targeted interventions (e.g., increasing study time or attendance)

However, about 33% of performance remains unexplained, indicating that: External factors (e.g., personal circumstances, teaching quality, exam difficulty) also matter.

Overall, regression models provide a moderately high and practically useful level of accuracy in predicting student exam performance, making them valuable tools for educational planning and decision-making, though they should be complemented with qualitative insights for a complete understanding.

5.2 Technique 2 (K-Means Clustering)

Objective

To apply K-Means clustering to segment students into meaningful groups based on their academic and behavioural characteristics.

Analytical Question

Can students be meaningfully segmented into distinct groups (e.g., high performers, average students, and at-risk students) based on their academic and behavioural characteristics?

Methodology:

To segment students into meaningful groups based on academic performance and behavioural characteristics, an unsupervised machine learning approach using K-Means clustering was implemented. This method groups students into clusters based on similarity across selected features.

A new variable, *Delta*, was created to capture the change in student performance by computing the difference between *Exam_Score* and *Previous_Scores*. This feature provides insight into improvement or declines in academic performance and enhances the clustering analysis.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df["Delta"] = df["Exam_Score"] - df["Previous_Scores"]
```

A subset of relevant variables was selected to represent both performance and behavior:

- Exam Score
- Performance change (*Delta*)
- Hours studied
- Attendance

These features were chosen to capture both current achievement and underlying study patterns.

Since K-Means clustering is sensitive to the scale of variables, all selected features were standardized using the StandardScaler technique. This ensures that each variable contributes equally to the clustering process and prevents features with larger magnitudes from dominating the results.

The Elbow Method was used to identify the optimal number of clusters (k). K-Means models were trained for a range of cluster values (k = 1 to 10), and the corresponding inertia (within-cluster sum of squares) was computed. The optimal k was selected at the point where the rate of decrease in inertia begins to slow, indicating diminishing returns from adding more clusters. Based on this analysis, $k = 3$ was chosen.

```
df_kmeans = df[['Exam_Score' , 'Delta' , 'Hours_Studied', 'Attendance']].copy()
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df_kmeans)

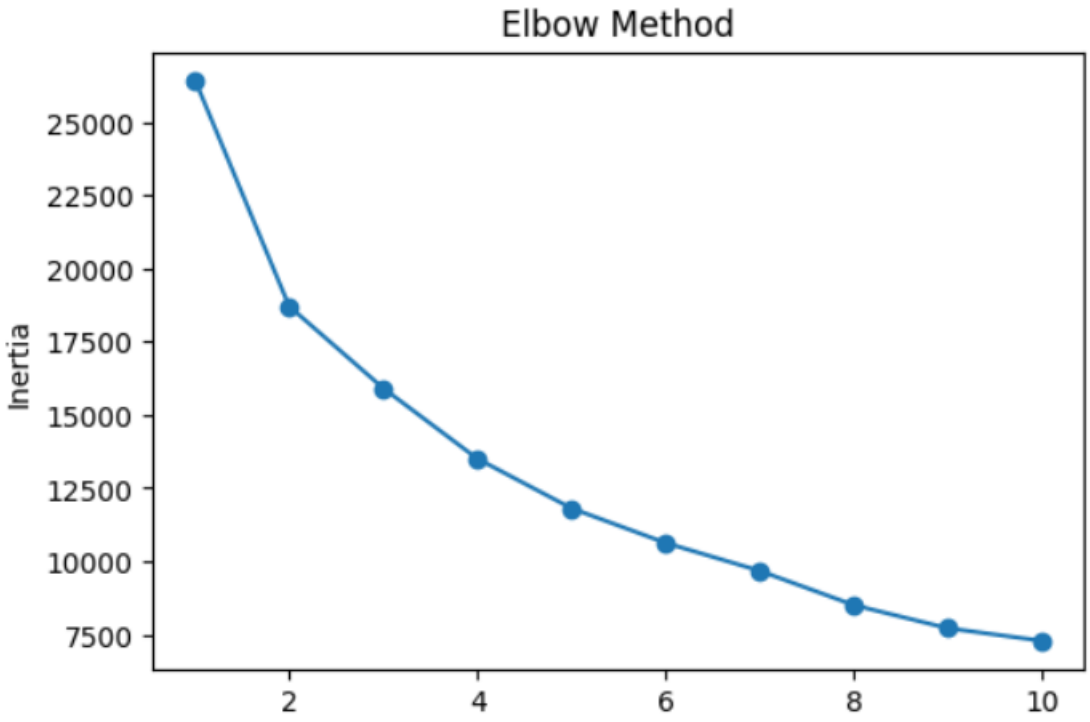
inertias = []

K_range = range(1, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    inertias.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(K_range, inertias, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.show()
```

The K-Means algorithm was applied with the selected number of clusters (k = 3) and a fixed random state to ensure reproducibility. Each student was assigned to a cluster based on similarity in the standardized feature space.



To interpret the clusters, group-wise averages of key variables (Exam Score, Delta, Hours Studied, and Attendance) were computed. These summaries provided insights into the characteristics of each cluster and allowed meaningful differentiation between student groups.

Clusters were further labelled into meaningful categories based on performance and improvement trends:

- **High Performing**: Students with high exam scores
- **Improving**: Students showing positive performance growth (high Delta)
- **At Risk**: Students with lower scores and/or declining performance
- **Average**: Students with moderate scores and stable performance

This labelling enhances interpretability and aligns the clusters with real-world educational insights.

```
import warnings
warnings.filterwarnings("ignore", category=DeprecationWarning,
module="jupyter_client")

optimal_k = 3

kmeans = KMeans(n_clusters=optimal_k, random_state=42)
df_kmeans["Cluster"] = kmeans.fit_predict(scaled_data)

cluster_summary = df_kmeans.groupby("Cluster")[["Exam_Score", "Delta",
"Hours_Studied", "Attendance"]].mean()

print("\nCluster Summary:")
print(cluster_summary)

# Rank clusters based on performance and trend
score_rank = cluster_summary["Exam_Score"].rank()
delta_rank = cluster_summary["Delta"].rank()

cluster_labels = {}

for cluster in cluster_summary.index:
    score = score_rank[cluster]
    delta = delta_rank[cluster]

    if score == max(score_rank) and delta < max(delta_rank):
        cluster_labels[cluster] = "High but Declining"
    elif delta == max(delta_rank):
        cluster_labels[cluster] = "Improving"
    elif delta == min(delta_rank):
        cluster_labels[cluster] = "At Risk"
    else:
        cluster_labels[cluster] = "Average"

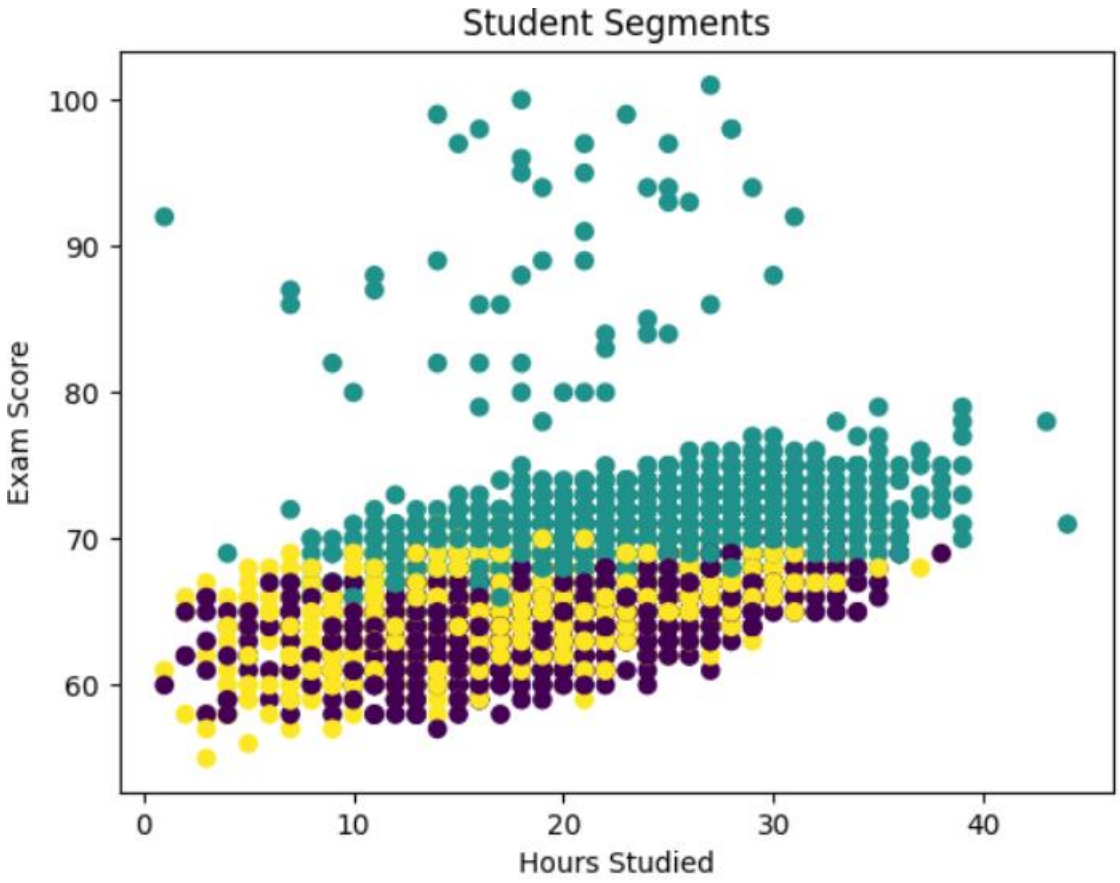
df_kmeans["Segment"] = df_kmeans["Cluster"].map(cluster_labels)

print("\nCluster Sizes:")
print(df_kmeans["Segment"].value_counts())

print("\nSegment Summary:")
print(df_kmeans.groupby("Segment")[["Exam_Score", "Delta", "Hours_Studied",
"Attendance"]].mean())

plt.figure()
plt.scatter(df_kmeans["Hours_Studied"], df_kmeans["Exam_Score"], c=df_kmeans
["Cluster"])
plt.xlabel("Hours Studied")
plt.ylabel("Exam Score")
plt.title("Student Segments")
plt.show()
```

A scatter plot was generated to visualize the clustering results, plotting Hours Studied against Exam Score and colouring data points by cluster membership. This visualization helps illustrate the separation between clusters and supports interpretation of student segments.



Results

```
Cluster Summary:
         Exam_Score      Delta  Hours_Studied  Attendance
Cluster
0         64.793156   2.999497      19.081530   73.707096
1         70.560427  -4.595972      22.225513   90.646919
2         65.528257 -22.072797      18.097222   73.006226

Cluster Sizes:
Segment
High but Declining    2532
At Risk               2088
Improving             1987
Name: count, dtype: int64

Segment Summary:
                    Exam_Score      Delta  Hours_Studied  Attendance
Segment
At Risk              65.528257 -22.072797      18.097222   73.006226
High but Declining   70.560427  -4.595972      22.225513   90.646919
Improving            64.793156   2.999497      19.081530   73.707096
```

Statistical Perspective

The Elbow Method indicates that the optimal number of clusters is three, as the inertia shows a sharp decline from k = 1 to k = 3 and then begins to level off, justifying the selection of three clusters. The clustering results segment students into three meaningful groups: At Risk, Improving, and High but Declining. At Risk students have moderate scores but show a significant negative trend in performance, lower study hours, and moderate attendance, indicating the need for immediate academic intervention. Improving students have similar scores but exhibit a positive performance trend, suggesting they are on the right track and can benefit from continued support. High but Declining students have the highest scores, study hours, and attendance, yet show a downward trend, highlighting the need for monitoring despite strong current performance. The cluster distribution reveals that a large proportion of students fall into either declining or at-risk categories, with fewer consistently improving. Additionally, the scatter plot shows a general positive relationship between study hours and exam scores, although overlaps between clusters suggest that performance is influenced by multiple factors beyond study time alone. Overall, the analysis highlights that student performance is dynamic, and the inclusion of performance trends (Delta) provides deeper insights, demonstrating that students with similar scores may follow very different trajectories, emphasizing the importance of targeted and timely interventions.

Practical Perspective

The clustering analysis enables early identification of students who are at risk of academic decline, allowing timely intervention before their performance worsens. Students classified as “At Risk” require immediate academic support and guidance, while those in the “Improving” category can benefit from continued encouragement and additional resources to sustain their progress. Furthermore, students identified as “High but Declining,” despite currently performing well, should be closely monitored to prevent potential deterioration in performance. Overall, this segmentation approach supports targeted and data-driven strategies, helping educators tailor interventions according to the specific needs and performance trends of different student groups.

The clustering analysis successfully segments students into three meaningful groups At Risk, Improving, and High but Declining based on academic performance and behavioural patterns. While study hours and attendance influence performance, the inclusion of performance trends (Delta) reveals deeper insights, highlighting that students with similar scores may follow very different trajectories. This segmentation provides a practical framework for targeted interventions, enabling educators to proactively support students based on their specific needs and performance trends.

5.3 Technique 3 (Decision Tree)

Objective

To assess how accurately at-risk students can be identified early using behavioural and academic indicators through predictive modelling techniques.

Analytical Question

How effectively can at-risk students be identified early using behavioural and academic indicators?

Methodology:

To assess how effectively at-risk students can be identified early, a supervised machine learning approach was adopted using a Decision Tree Classifier. This method enables the classification of students into "at-risk", and "not at-risk" categories based on academic and behavioural indicators.

A binary target variable, *At_Risk*, was created from the *Previous_Scores* feature. Students with scores below a defined threshold (e.g., 60) were labelled as "at-risk" (1), while others were labelled as "not at-risk" (0). This transformation converts the problem into a classification task.

Relevant predictor variables were selected to capture both academic and behavioral aspects of student performance. These included:

- Hours studied
- Attendance
- Sleep hours
- Motivation level
- Tutoring sessions
- Physical activity

These features were chosen based on their potential influence on academic outcomes.

Categorical variables within the feature set were encoded into numerical format using one-hot encoding to ensure compatibility with the machine learning model. To avoid multicollinearity, one category from each encoded variable was dropped.

The dataset was divided into training and testing subsets using an 80:20 split. The training set was used to build the model, while the testing set was used to evaluate its predictive performance. A fixed random state was applied to ensure reproducibility.

A Decision Tree Classifier was implemented using the entropy criterion, which measures information gain to determine optimal splits at each node. The tree depth was limited to a maximum of three levels to control model complexity and reduce the risk of overfitting, while maintaining interpretability.

The model was trained on the training dataset, learning decision rules that partition students based on feature values. These learned rules were then applied to the test dataset to predict whether each student falls into the at-risk category.

Model performance was evaluated using classification accuracy, which measures the proportion of correctly classified instances in the test set. The obtained accuracy score (approximately 0.81) indicates that the model can correctly classify a substantial proportion of students.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn import metrics
import pandas as pd

# Create label
df["At_Risk"] = df["At_Risk"] = df["Previous_Scores"].apply(lambda x: 1 if x <
60 else 0)

# Features
feature_cols = ["Hours_Studied", "Attendance", "Sleep_Hours",
                "Motivation_Level","Tutoring_Sessions","Physical_Activity"]

X = df[feature_cols]

# FIX 1: encode categorical
X = pd.get_dummies(X, drop_first=True)

y = df["At_Risk"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Model
dtree = DecisionTreeClassifier(criterion='entropy', max_depth=3)
model = dtree.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

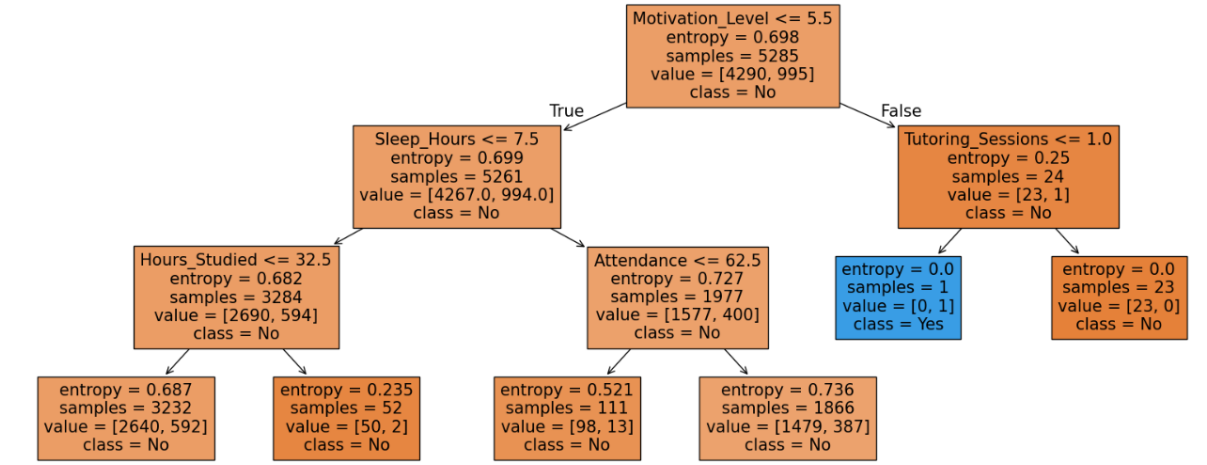
# Evaluate
print("Accuracy:", metrics.accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.8184568835098336
```

To enhance interpretability, the trained decision tree was visualized. This graphical representation illustrates the hierarchical decision-making process, showing how different features and thresholds contribute to classification outcomes. Such visualization aids in understanding key risk factors and supports explainable decision-making.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(20, 8))
plot_tree(model,
          feature_names=feature_cols,
          class_names=["No", "Yes"],
          filled=True)
plt.show()
```



Statistical Perspective

The Decision Tree Classifier achieved an accuracy of approximately 81.85%, indicating that the model correctly classifies students as “At Risk” or “Not at Risk” in about 4 out of 5 cases. This reflects a strong predictive performance for a classification problem in an educational context.

The tree structure shows that Motivation Level is the most important splitting variable (root node), followed by factors such as Sleep Hours, Study Hours, Attendance, and Tutoring Sessions. These variables significantly contribute to distinguishing between at-risk and non-risk students.

Most nodes are dominated by the “Not at Risk” class, suggesting a class imbalance, where fewer students fall into the at-risk category. The entropy values decrease as the tree progresses, indicating improved classification purity at deeper levels.

Practical Perspective

From a real-world perspective, the model highlights that student motivation is the primary driver in identifying at-risk students. Students with lower motivation levels are more likely to be at risk, especially when combined with lower sleep duration, fewer study hours, and poor attendance.

Additionally:

Students with low attendance and low study hours are more likely to fall into the at-risk category. Tutoring sessions appear to reduce risk, as students receiving support are less likely to be classified as at risk. Adequate sleep and consistent study habits contribute positively to reducing academic risk.

This model allows educators to identify vulnerable students early using simple and observable behavioural indicators, enabling timely intervention.

The Decision Tree model demonstrates strong performance in identifying at-risk students with good accuracy and interpretability. It reveals that behavioural and academic factors particularly motivation, study habits, attendance, and sleep play a crucial role in determining student risk levels. The model provides a practical and actionable framework for early detection and targeted intervention, helping educators proactively support students and improve academic outcomes.

6. Recommendations

Based on the analytical findings from regression, clustering, and classification models, the following recommendations are proposed:

Implement Early Warning Systems

Early warning systems should be used to identify at-risk students based on behavioural and academic indicators such as low motivation, attendance, and study hours. This enables timely intervention before performance declines significantly. Proactive identification helps educators provide targeted support at the right time.

Strengthen Study and Attendance Interventions

Since study hours and attendance are key predictors of performance, institutions should actively monitor and improve these factors. Setting minimum benchmarks and tracking engagement can help ensure consistency. Even small improvements in these areas can lead to meaningful academic gains.

Expand Access to Tutoring and Academic Support

Tutoring has a strong positive impact on student outcomes, making it essential to expand access to academic support services. Special focus should be given to at-risk and declining students. Structured support programs can help bridge learning gaps and improve performance.

Encourage Greater Parental Involvement

Higher levels of parental involvement are consistently associated with better academic performance. Schools should promote active parent engagement through regular communication and participation in academic planning. This creates a supportive learning environment beyond the classroom.

Use Segmentation for Targeted Strategies

Student segmentation allows educators to categorize students into groups such as at-risk, improving, and high but declining. This enables more personalized and effective intervention strategies. Tailored approaches ensure that each student group receives the support they specifically need.

Improve Access to Learning Resources

Access to academic and digital resources plays a crucial role in student success. Ensuring equitable availability of learning materials and internet access can reduce performance gaps. Improved resource accessibility supports consistent and effective learning.

7. Summary

This study demonstrates that student performance can be effectively analysed, predicted, and segmented using a combination of statistical and machine learning techniques. The regression model explains approximately 67% of the variation in exam scores, confirming that academic behaviours such as study hours, attendance, and tutoring are key drivers of success.

Clustering analysis further reveals that students follow distinct performance trajectories, including at-risk, improving, and high-but-declining groups, highlighting that performance is dynamic rather than static. Additionally, the decision tree model achieves an accuracy of over 81%, showing that at-risk students can be identified early using observable behavioural indicators, with motivation emerging as the most influential factor.

Overall, the findings emphasize that student success is shaped by a combination of academic effort, behavioural patterns, and environmental support. By leveraging data-driven insights, educators can move from reactive to proactive intervention strategies, enabling timely support, personalized learning approaches, and improved academic outcomes.